

Notes: Inoue and Kilian (JPE, 2026)

When Is the Use of Gaussian-Inverse Wishart-Haar Priors Appropriate?

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1 Big picture

- **Question.** When is the conventional Bayesian approach to sign-identified VARs—Gaussian-inverse Wishart priors for reduced-form parameters plus a Haar prior for the rotation matrix—a reasonable basis for posterior inference about impulse responses?
- **Setting.** The paper is mostly theoretical and methodological. It combines a stylized set-identified model, a geometric bivariate VAR example, and three empirical illustrations from the recent structural VAR literature.
- **Core challenge.** In sign-identified VARs, the likelihood is flat over the identified set. Asymptotically, posterior inference over that set is therefore driven by the prior on the rotation matrix Q . Recent critiques conclude from this that conventional Bayesian impulse-response inference is largely prior driven.
- **Main conceptual contribution.** The paper argues that the quantitative importance of the Haar prior has been overstated. What matters is not simply that the posterior over the identified set depends on the prior for Q , but whether the identified set is wide enough for that dependence to matter economically.
- **Main mechanism.** The sensitivity of posterior inference to the Haar prior depends on the *width of the identified set*. A narrow identified set can make conventional posterior means and credible sets practically indistinguishable from prior-robust alternatives, even though the prior still shapes the distribution inside the set.
- **Why this is subtle.** The width of the identified set is determined not only by the number and type of identifying restrictions, but also by the data through the reduced-form parameters (A, Σ) . Hence there is no universal statement that the Haar prior is or is not problematic.
- **Main diagnostic idea.** Compare conventional Bayes estimates and credible intervals to prior-robust identified sets and credible regions using the *Hausdorff distance*. If the distance is economically negligible, the Haar prior can be ignored for practical purposes.
- **Main theoretical result.** In a stylized Watson (2020)-type set-identification problem, prior sensitivity is dramatic when the identified set is wide and economically irrelevant when the set is sufficiently narrow.
- **Main empirical result.** The paper shows both kinds of cases. In weakly identified models such as Uhlig (2005), the conventional approach is problematic. In more tightly identified models with additional sign, narrative, exclusion, or other restrictions, conventional inference can be very close to prior-robust inference.
- **Economic takeaway.** The right question is not “Does the Haar prior matter in principle?” The right question is “How much does it matter in the application at hand?” The answer depends on how tight identification is given both the restrictions and the data.

2 Data and measurement (key objects)

2.1 Core objects in sign-identified VARs

The paper studies structural VARs in which:

- A denotes the reduced-form slope parameters,
- Σ denotes the reduced-form covariance matrix,
- Q is an orthogonal rotation matrix,
- and the scalar impulse response of interest is

$$v = g(A, \Sigma, Q). \quad (1)$$

The conventional prior factorizes as

$$p(A, \Sigma, Q) = p(A, \Sigma) p(Q), \quad (2)$$

where $p(A, \Sigma)$ is Gaussian-inverse Wishart and $p(Q)$ is Haar.

Interpretation. The paper is not about whether the Haar prior enters the posterior—it does. It is about whether the resulting influence on economically relevant posterior summaries is material.

2.2 The stylized Watson-type model

To clarify the logic, the paper studies a minimal set-identified model in which the observed data satisfy

$$Y \mid m \sim N(m, I_2/n), \quad (3)$$

with $Y = (Y_1, Y_2)'$ and $m = (m_1, m_2)'$, under the support restriction $m_1 < m_2$.

The scalar parameter of interest v is set identified by the sign restriction

$$m_1 \leq v \leq m_2. \quad (4)$$

Because the likelihood does not depend directly on v , the posterior is

$$p(v \mid Y) = \int_{m_1 < m_2} p(v \mid m_1, m_2) p(m_1, m_2 \mid Y) d(m_1, m_2). \quad (5)$$

Interpretation. This model strips away all VAR machinery and isolates the key issue: posterior sensitivity for a set-identified scalar depends on the width of the admissible interval $[m_1, m_2]$.

2.3 Alternative priors over the set-identified parameter

Conditional on (m_1, m_2) , the paper considers four priors for v :

$$\text{Prior 1: } v \mid m \sim U(m_1, m_2), \quad (6)$$

$$\text{Prior 2: } v \mid m \sim N(0, 1) \text{ truncated to } [m_1, m_2], \quad (7)$$

$$\text{Prior 3: } P(v = m_1 \mid m) = 0.99, P(v = m_2 \mid m) = 0.01, \quad (8)$$

$$\text{Prior 4: } P(v = m_1 \mid m) = 0.01, P(v = m_2 \mid m) = 0.99. \quad (9)$$

Interpretation. Priors 3 and 4 are intentionally extreme. They show that even wildly different posterior weights inside the identified set can be economically harmless if the set is tight enough.

2.4 Posterior estimate of the identified set

The paper defines the posterior estimate of the identified set as

$$\hat{\mathcal{I}}(Y) = [E(m_1 | Y), E(m_2 | Y)]. \quad (10)$$

If $\bar{v}(Y)$ is the posterior mean of v under a given prior, the maximum error from reporting $\bar{v}(Y)$ instead of the set estimate is

$$\max\{|\bar{v}(Y) - E(m_1 | Y)|, |\bar{v}(Y) - E(m_2 | Y)|\}. \quad (11)$$

Interpretation. This is the one-dimensional version of the Hausdorff distance between a point estimate and an interval estimate.

2.5 Hausdorff distance and probability bounds

For two intervals $[a, b]$ and $[c, d]$, the Hausdorff distance is

$$d_H([a, b], [c, d]) = \max\{|a - c|, |b - d|\}. \quad (12)$$

The paper also considers the identified set of posterior probabilities such as $P(v \geq 0 | Y)$. When $m_1 \leq 0 \leq m_2$, finite-sample bounds can be informative even though asymptotically they may widen to $[0, 1]$.

A convenient representation of those bounds is

$$\mathcal{P}_n(Y) = [\underline{P}_n(Y), \bar{P}_n(Y)], \quad (13)$$

where, in the Watson model,

$$\underline{P}_n(Y) = \int_0^\infty \frac{\sqrt{n}}{c} \phi(\sqrt{n}(m_1 - Y_1)) \Phi(-\sqrt{n}(m_1 - Y_2)) dm_1, \quad (14)$$

$$\bar{P}_n(Y) = \int_0^\infty \frac{\sqrt{n}}{c} \phi(\sqrt{n}(m_2 - Y_2)) \Phi(\sqrt{n}(m_2 - Y_1)) dm_2. \quad (15)$$

Here ϕ and Φ are the standard normal density and cdf and c is a normalizing constant.

Interpretation. A tight identified set can make even posterior sign probabilities fairly informative at finite sample sizes.

2.6 The empirical objects in the illustrations

The paper's applied examples use three kinds of quantities:

- **estimated identified sets** for impulse responses,

- **conventional Bayes estimates** and conventional credible intervals under the Gaussian-inverse Wishart-Haar prior,
- and **prior-robust credible intervals or probability bounds** that do not take a stand on the prior for Q .

Interpretation. The whole paper is about comparing these objects and asking whether the differences are economically meaningful.

3 Models and empirical specifications (all equations + interpretation)

3.1 The conventional sign-identified VAR setup

The conventional Bayesian approach places a conjugate prior on (A, Σ) and a Haar prior on Q , then maps these objects into impulse responses via $v = g(A, \Sigma, Q)$.

The crucial asymptotic point is that, once the reduced-form parameters are point identified from the data, the likelihood is flat over the identified set for v . Asymptotically, posterior inference inside the identified set is therefore driven by the prior for Q .

Interpretation. This is the logical basis of the critique in Baumeister and Hamilton (2015), Plagborg-Møller (2019), Watson (2020), and Giacomini–Kitagawa (2021).

3.2 Why the width of the identified set is the decisive object

The paper’s central claim is simple:

- if the identified set is *wide*, the prior-induced posterior distribution inside the set can substantially change posterior means and credible sets;
- if the identified set is *narrow*, the prior still matters formally, but it matters little economically because all admissible responses are close to one another.

In the stylized model this difference is transparent:

- when $(Y_1, Y_2) = (-2.5, 2.5)$, the identified set is wide and priors generate very different posterior summaries;
- when $(Y_1, Y_2) = (-0.05, 0.05)$, the identified set is narrow and even extreme priors yield economically similar posterior summaries.

3.3 A bivariate VAR illustration of data dependence

To show why set width depends on the data, the paper uses a bivariate VAR(0) model of price and quantity:

$$\begin{pmatrix} P_t \\ Q_t \end{pmatrix} = B_0^{-1} \begin{pmatrix} \omega_t^S \\ \omega_t^D \end{pmatrix}, \quad (16)$$

with structural impact matrix

$$B_0^{-1} = \begin{pmatrix} \cos(\nu) & \sin(\nu) \\ \sin(\Theta + \nu) & -\cos(\Theta + \nu) \end{pmatrix}, \quad (17)$$

and reduced-form correlation

$$r = \sin(\Theta). \quad (18)$$

Supply and demand shocks are identified using sign restrictions on the impact matrix entries. The width of the identified set for the response of quantity to a demand shock changes sharply with r , even though the sign restrictions are held fixed.

Interpretation. Identification tightness is not solely about the identifying assumptions. It also depends on the realized reduced-form covariance structure.

3.4 The practical diagnostic for applied work

The paper proposes a simple decision rule for applied users.

First, estimate the identified set of the impulse response without taking a stand on the prior for Q . Conceptually, this can be represented as

$$\hat{\mathcal{I}}_h = \left[\frac{1}{S} \sum_{s=1}^S \min_{Q \in \mathcal{Q}_s} g_h(A^{(s)}, \Sigma^{(s)}, Q), \frac{1}{S} \sum_{s=1}^S \max_{Q \in \mathcal{Q}_s} g_h(A^{(s)}, \Sigma^{(s)}, Q) \right], \quad (19)$$

where h denotes the horizon and $\mathcal{Q}_s$ is the set of admissible rotations for reduced-form draw s .

Second, compare the conventional Bayes estimate under the Haar prior, say $\hat{v}_h^H$, to the set estimate. The relevant metric is

$$d_H(\hat{v}_h^H, \hat{\mathcal{I}}_h) = \max\{|\hat{v}_h^H - \underline{I}_h|, |\hat{v}_h^H - \bar{I}_h|\}. \quad (20)$$

Third, compare the conventional credible interval $[L_h^H, U_h^H]$ to the robust interval $[L_h^R, U_h^R]$ using

$$d_H([L_h^H, U_h^H], [L_h^R, U_h^R]) = \max\{|L_h^H - L_h^R|, |U_h^H - U_h^R|\}. \quad (21)$$

Interpretation. The conventional approach is acceptable when these distances are economically negligible, not necessarily numerically zero.

3.5 Why the Hausdorff metric is attractive

The paper emphasizes two advantages:

1. the Hausdorff distance is measured in the same units as the impulse response, making it easy to interpret economically;
2. it reflects both the *width* and the *location* of the identified set relative to the conventional estimate.

Interpretation. A narrow set far away from the conventional estimate is a problem, and a wide set centered on the conventional estimate is also a problem. The Hausdorff metric captures both.

3.6 What the empirical illustrations actually estimate

The empirical section does not estimate a new structural model. Instead, it re-examines well-known applications and asks three practical questions:

- How wide is the estimated identified set?
- How far is the conventional posterior mean from the set estimate?
- How far are conventional credible intervals from prior-robust credible intervals?

Interpretation. The paper is a diagnostic framework for reading and evaluating sign-identified VAR results.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no IV in the usual sense

This paper does *not* use an instrumental-variable strategy. Credibility comes instead from analytical identification and transparent comparisons between conventional and prior-robust objects.

Interpretation. The key identifying variation is geometric and probabilistic rather than quasi-experimental.

4.2 What identifies the paper’s main claim

The main claim—that the importance of the Haar prior is case specific—is identified by three ingredients:

1. the set-identifying restrictions,
2. the reduced-form parameters (A, Σ) learned from the data,
3. and direct comparison between conventional and robust posterior summaries.

If these comparisons are small in one application and large in another, the claim that prior sensitivity varies across applications follows directly.

4.3 Why the critique cannot be universal

A universal critique would require the width of the identified set to depend only on the prior for Q or only on the identifying restrictions. The paper shows neither is true:

- in the stylized model, width depends on how tightly the data pin down (m_1, m_2) ;
- in sign-identified VARs, width depends on the covariance structure Σ and, more generally, the reduced-form slope parameters A ;
- additional narrative, shape, exclusion, or elasticity restrictions can sharply narrow the admissible

set.

Interpretation. Whether the Haar prior is practically important is an empirical feature of a model-data combination, not a theorem about all sign-identified VARs.

4.4 Why reverse logic from asymptotics can mislead

The literature often stresses that, asymptotically, the posterior over the identified set is determined by the prior for Q . The paper’s point is that this statement by itself does not tell us how large the quantitative distortion will be in any application.

Interpretation. “Prior driven in principle” does not necessarily mean “economically misleading in practice.”

4.5 Why the finite-sample perspective matters

Even when asymptotically the identified set for a sign probability can reach $[0, 1]$, finite-sample bounds can still be informative if the identified set is narrow. Likewise, conventional posterior means may be close enough to robust identified-set bounds to be adequate summaries for applied work.

Interpretation. The paper is a reminder not to jump from asymptotic logic to blanket empirical pessimism.

4.6 What the paper can and cannot distinguish

The paper can distinguish:

- cases in which conventional inference is clearly unreliable because identified sets are wide,
- cases in which conventional inference is practically reliable because identified sets are narrow.

The paper cannot fully rank all possible priors or resolve the deeper philosophical question of what prior over Q is normatively correct.

Interpretation. The paper is pragmatic. It does not defend the Haar prior as universally right; it defends its use when diagnostics show that it does not matter much.

4.7 Relation to the literature

Relative to Baumeister and Hamilton (2015), Watson (2020), and Giacomini–Kitagawa (2021), this paper does not dispute that the Haar prior shapes the posterior over a set. Instead, it shifts the question from *whether* this happens to *how much it matters quantitatively* in actual applications.

Interpretation. The contribution is not a denial of the critique; it is a calibration of the critique.

5 Tables and figures

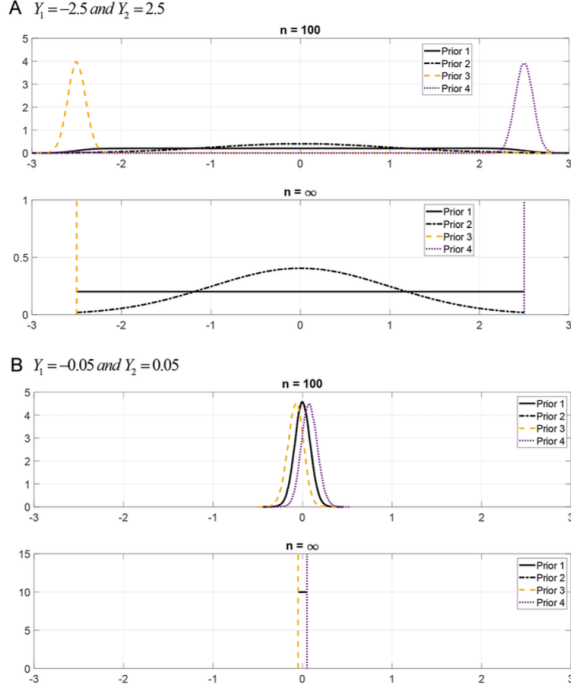


FIG. 1.—Posterior distribution of θ in Watson (2020) model under alternative priors given wide (A) and narrow (B) identified sets. For $n = 100$, we estimate the posterior densities using a kernel density smoother. The densities for $n = \infty$ are derived analytically. The posterior under priors 3 and 4 assigns 0.99 probability mass to one endpoint of the identified set and 0.01 to the other. For simplicity, we represent this result as a vertical line at the endpoint with 0.99 probability mass.

	$Y_1 = -2.5$ and $Y_2 = 2.5$				$Y_1 = -0.05$ and $Y_2 = 0.05$			
	$n = 100$	$n = \infty$	$n = 100$	$n = \infty$	$n = 100$	$n = \infty$	$n = 100$	$n = \infty$
	Posterior Mean (1)	Maximum Error (2)	Posterior Mean (3)	Maximum Error (4)	Posterior Mean (5)	Maximum Error (6)	Posterior Mean (7)	Maximum Error (8)
Prior 1	-.00	2.50	.00	2.50	.00	.08	.00	.05
Prior 2	.00	2.50	.00	2.50	.00	.08	-.00	.05
Prior 3	-2.45	4.95	-2.45	4.95	-.08	.16	-.05	.10
Prior 4	2.45	4.95	2.45	4.95	.08	.16	.05	.10

NOTE.—The Watson (2020) model is a stylized model with a generic set-identified parameter. The structure of the model and priors is discussed in the main text. The maximum error is the Hausdorff distance between the posterior mean and the posterior estimate of the identified set shown in table A1.

5.1 Figure 1 and Table 1: the stylized Watson example

Read:

- Figure 1 compares posterior distributions of v under four priors when the identified set is wide, $[-2.5, 2.5]$, and when it is narrow, $[-0.05, 0.05]$.
- Table 1 shows that with the wide set, the maximum error from reporting the posterior mean can be as large as 4.95 under endpoint-heavy priors.
- With the narrow set, the maximum error collapses to 0.05 under diffuse priors and to 0.10 under the extreme endpoint priors.

Economic message. The same formal dependence on the prior can be either economically disastrous or economically negligible, depending on the width of the identified set.

5.2 Table 2: probability bounds in the stylized model

Read:

- When the identified set is wide, the identified set of $P(v \geq 0 \mid Y)$ is essentially $[0, 1]$ for all n considered.
- When the identified set is narrow, the bounds are informative in finite samples: for $n = 10$ they are $[0.23, 0.77]$ and for $n = 100$ they are $[0.19, 0.81]$.

- As $n \rightarrow \infty$, the bounds return to $[0, 1]$ when zero lies in the identified set.

Economic message. Posterior sign probabilities can still be informative at finite samples, but their informativeness is again tied to how tightly the parameter is identified.

TABLE 2
IDENTIFIED SET OF $P(\theta \geq 0|Y)$ IN WATSON (2020) MODEL

Y_1, Y_2	Identified Set of $P(\theta \geq 0 Y)$					
	$n = 10$		$n = 100$		$n = \infty$	
± 2.5	.00	1.00	.00	1.00	.00	1.00
$\pm .05$	.23	.77	.19	.81	.00	1.00

NOTE.—Entries are obtained using the expression for the interval endpoints derived in the appendix.

TABLE 3
POSTERIOR RESPONSE OF QUANTITY TO DEMAND SHOCK IN UHLIG (2017) MODEL FOR $n = \infty$

ρ	$U(0, 2\pi)$			$2\pi\text{Beta}(2.5, 5)$			$2\pi\text{Beta}(5, 2.5)$		
	90% Credible Interval			90% Credible Interval			90% Credible Interval		
	Identified Set			Identified Set			Identified Set		
	Lower Bound (1)	Upper Bound (2)	Mean (3)	Lower Bound (4)	Upper Bound (5)	Mean (6)	Lower Bound (7)	Upper Bound (8)	Mean (9)
−.999	.000	.045	.022	.002	.042	.022	.002	.042	.022
−.990	.000	.141	.071	.007	.134	.069	.007	.134	.072
−.900	.000	.436	.222	.023	.415	.211	.020	.412	.237
−.500	.000	.866	.477	.052	.839	.439	.043	.828	.560
.000	.000	1.000	.637	.079	.997	.590	.067	.993	.794

NOTE.—The table is based on the bivariate model of prices and quantities with sign restrictions on the response to demand and supply shocks. Because $n = \infty$, the posterior over the identified set equals the prior.

5.3 Table 3: reduced-form correlation and identified-set width

Read:

- In the Uhlig (2017) bivariate price-quantity example, when the reduced-form correlation is $r = 0$, the identified set for the quantity response to a demand shock is $[0, 1]$.
- When $r = -0.5$, the width falls to 0.866.
- When $r = -0.999$, the width collapses to 0.045.
- Under one prior, the posterior mean ranges from 0.022 to 0.794 as r changes.

Economic message. Data matter directly for how much the prior matters. Holding the sign restrictions fixed, different reduced-form covariance structures can make the model either tightly or weakly identified.

5.4 Tables 4 and 5: Baumeister–Hamilton monetary-policy example

TABLE 4
POSTERIOR MEAN OF IMPACT RESPONSE OF OUTPUT GAP TO MONETARY POLICY TIGHTENING

Model	Estimate of Identified Set			Conventional Bayes Estimate	
	Lower Bound	Upper Bound	Width	Posterior Mean	Maximum Error
No sign restrictions	−.482	.482	.964	−.005	.487
Impact sign restrictions	−.394	−.000	.394	−.111	.284
Impact + shock sign restrictions	−.088	−.000	.088	−.030	.059

NOTE.—The table is based on the same reduced-form VAR model as in Baumeister and Hamilton (2018). The estimates of the identified set were computed as in Giacomini, Kitagawa, and Read (2023). The shock is a 25-bp increase in the policy rate.

TABLE 5
68% CREDIBLE INTERVAL FOR IMPACT RESPONSE OF OUTPUT GAP TO MONETARY POLICY TIGHTENING

Model	Robust Credible Interval			Conventional Credible Interval		
	Lower Bound	Upper Bound	Width	Lower Bound	Upper Bound	Maximum Error
No sign restrictions	−.499	.501	1.000	−.298	.346	.643
Impact sign restrictions	−.411	−.000	.411	−.142	−.000	.142
Impact + shock sign restrictions	−.109	.000	.109	−.033	.000	.033

NOTE.—See table 4.

Read:

- With no sign restrictions, the identified set for the impact response of the output gap to a 25 bp monetary tightening is $[-0.482, 0.482]$, with width 0.964 and maximum error of the posterior mean equal to 0.487.

- Adding impact sign restrictions narrows the set to width 0.394 and reduces the maximum error to 0.284.
- Adding both impact sign restrictions and narrative shock-sign restrictions narrows the set to width 0.088 and the maximum error to 0.059.
- Table 5 shows the same pattern for 68% credible intervals: the maximum error falls from 0.202 to 0.076.

Economic message. The same reduced-form model can move from “the Haar prior is a serious problem” to “the Haar prior is practically irrelevant” once the identification is made sufficiently informative.

5.5 Tables 6, 7, and 8: Uhlig (2005) versus Antolin-Díaz–Rubio-Ramírez (2018)

TABLE 6
POSTERIOR MEAN OF GDP RESPONSE TO MONETARY POLICY TIGHTENING IN UHLIG (2005) MODEL

Horizon (months)	Estimate of Identified Set			Conventional Bayes Estimate	
	Lower Bound	Upper Bound	Width	Posterior Mean	Maximum Error
12	-.226	.363	.589	.116	.342
24	-.299	.312	.611	.039	.338
36	-.318	.333	.651	.041	.358
48	-.309	.330	.648	.043	.352
60	-.299	.323	.622	.035	.333

NOTE.—The static and dynamic sign restrictions are described in Uhlig (2005). The shock is a 25-bp increase in the policy rate. The responses are in percentages.

TABLE 7
RANGE OF POSTERIOR PROBABILITIES OF DECLINE IN GDP (%)

Horizon (months)	Uhlig (2005) Model		Antolin-Díaz and Rubio-Ramírez (2018) Model	
	Minimum (1)	Maximum (2)	Minimum (3)	Maximum (4)
12	.0	100.0	15.2	96.9
24	.0	100.0	78.9	100.0
36	.0	100.0	72.0	100.0
48	.0	100.0	63.8	99.6
60	.6	99.9	62.0	99.2

NOTE.—The static and dynamic sign restrictions are described in Uhlig (2005). The Antolin-Díaz and Rubio-Ramírez (2018) model includes additional narrative restrictions on the shock sign and on the relative importance of policy shocks for explaining the federal funds rate in selected periods.

TABLE 8
POSTERIOR MEAN GDP RESPONSE TO MONETARY POLICY TIGHTENING IN UHLIG (2005) MODEL WITH ADDED NARRATIVE RESTRICTIONS 6 AND 7 AS IN ANTOLIN-DÍAZ AND RUBIO-RAMÍREZ (2018)

Horizon (months)	Estimate of Identified Set			Conventional Bayes Estimate	
	Lower Bound	Upper Bound	Width	Posterior Mean	Maximum Error
12	-.128	.082	.211	-.021	.107
24	-.310	-.070	.240	-.205	.133
36	-.332	-.060	.272	-.214	.154
48	-.317	-.042	.275	-.197	.155
60	-.298	-.035	.263	-.184	.149

NOTE.—The two sets of additional narrative restrictions include shock sign restrictions and restrictions on the relative importance of policy shocks for explaining the federal funds rate in selected periods, as described in Antolin-Díaz and Rubio-Ramírez (2018). The responses are in percentages.

Read:

- In Uhlig (2005), the response of GDP to a 25 bp monetary tightening has identified-set widths between 0.589 and 0.651 at horizons 12–60 months.
- The corresponding maximum errors of the posterior mean lie between 0.333 and 0.358.
- Table 7 shows that the posterior probability of a GDP decline ranges from 0% to 100% at most horizons: the model is essentially uninformative about the sign.
- After adding narrative restrictions as in Antolin-Díaz and Rubio-Ramírez (2018), widths fall to about 0.211–0.275 and maximum errors to about 0.107–0.155.
- At a two-year horizon, the probability of a GDP decline becomes bounded between 78.9% and 100%.

Economic message. This is the paper’s most balanced example. The conventional approach is weak in the original agnostic model, but becomes fairly accurate once narrative restrictions sharpen identification.

5.6 Table 9: Kilian–Zhou gasoline-price shock example

Read:

- For the impact response of the real gasoline price, the identified set is [3.874, 4.396] with posterior mean 4.231 and maximum error 0.357.

TABLE 9
RESPONSES TO NOMINAL GASOLINE PRICE SHOCK IN KILIAN AND ZHOU (2022) MODEL

Horizon (months)	Estimate of Identified Set			Conventional Bayes Estimate	
	Lower Bound	Upper Bound	Width	Posterior Mean	Maximum Error
A. Real Gasoline Price					
0	3.874	4.396	.522	4.231	.357
1	5.796	6.477	.681	6.276	.479
2	5.866	6.493	.627	6.312	.446
3	5.453	5.885	.432	5.770	.317
4	5.218	5.552	.334	5.466	.248
5	4.732	5.160	.428	5.038	.306
B. Headline Inflation					
0	.174	.213	.038	.197	.023
1	.092	.105	.013	.100	.008
2	.006	.014	.009	.010	.004
3	-.012	-.004	.008	-.008	.004
4	.004	.008	.005	.006	.002
5	-.006	.011	.017	.003	.009
C. Expected Inflation					
0	.160	.182	.022	.174	.014
1	.226	.246	.020	.240	.014
2	.217	.239	.022	.232	.014
3	.163	.197	.035	.183	.021
4	.152	.195	.043	.177	.025
5	.115	.138	.023	.129	.013

NOTE.—The model is identified on the basis of sign and exclusion restrictions on the structural impact multiplier matrix, as described in Kilian and Zhou (2022). The responses are in percentages.

- For headline inflation, the impact identified set is $[0.174, 0.213]$ with posterior mean 0.197 and maximum error 0.023.
- For expected inflation, the impact identified set is $[0.160, 0.182]$ with posterior mean 0.174 and maximum error 0.014.

Economic message. Even without narrative restrictions, some models are tightly enough identified that the conventional posterior is essentially data driven. This example is the clearest positive case for the Gaussian-inverse Wishart-Haar approach.

6 Short summary

- The paper asks when conventional Bayesian inference in sign-identified VARs is genuinely distorted by the Haar prior on the rotation matrix.
- Its answer is: the distortion depends on the width of the identified set, and that width depends on both the identifying restrictions and the reduced-form data.
- In a stylized set-identified model, the paper shows that wide identified sets produce strong prior sensitivity while narrow identified sets make even extreme priors practically irrelevant.
- In sign-identified VARs, the paper proposes using the Hausdorff distance between conventional estimates and prior-robust identified sets or credible intervals as a practical diagnostic.
- Applications to monetary-policy and gasoline-price VARs show both kinds of cases: some models are clearly too weakly identified for conventional inference, while others are tight enough that the Gaussian-inverse Wishart-Haar prior is harmless in practice.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the debate over the Haar prior cannot be settled with a blanket statement. Conventional posterior inference in sign-identified VARs is not automatically spurious, but it is not automatically reliable either. Its usefulness depends on how tightly the object of interest is identified in the application at hand.

7.2 Economic intuition

When the identified set is wide, a prior over the rotation matrix determines where inside that wide region posterior mass is placed, and posterior means or credible sets can then be highly misleading. When the identified set is narrow, the data and identifying restrictions have already restricted the economically relevant possibilities so sharply that different priors over Q imply nearly the same substantive conclusion.

7.3 Takeaway

The paper's practical message is highly useful for empirical macroeconomists: do not abandon conventional sign-identified VAR methods mechanically, but do not trust them mechanically either. First estimate how wide the identified set is, compare conventional summaries to prior-robust ones, and then judge whether the Hausdorff distance is economically meaningful. In this paper's view, that is the right way to decide when the Gaussian-inverse Wishart-Haar prior is appropriate.